SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

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QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5

9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	**5**
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	**5**

Code of Conduct:

I __K.SANDHYA 192424061__________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

QUESTION : 1. STUDENT_COURSE — Functional Dependencies and Candidate Keys

Relation:

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Assume:

- One student has one name.
- One course has one course name and one instructor.

- A student's grade depends on the student and course together.

Functional Dependencies

1. SID → SName
2. CID → CName
3. CID → Instructor
4. SID, CID → Grade

Therefore:

SID → SName

CID → CName, Instructor

SID CID → Grade

Candidate Key

To determine all attributes, we need both SID and CID.

Closure:

(SID, CID)+

= {SID, CID}

Using SID → SName:

= {SID, CID, SName}

Using CID → CName, Instructor:

= {SID, CID, SName, CName, Instructor}

Using SID,CID → Grade:

= {SID, CID, SName, CName, Instructor, Grade}

Therefore:

Candidate Key = {SID, CID}

QUESTION : 2. Converting an Unnormalized Relation into 1NF

Example

Suppose we have:

STUDENT(SID, SName, PhoneNumbers)

SID	SName	PhoneNumbers
S1	Ravi	9876, 8765
S2	Priya	7654

The PhoneNumbers attribute contains multiple values.

This violates 1NF because every attribute must contain a single atomic value.

Step 1: Identify repeating groups

For student S1:

PhoneNumbers = 9876, 8765

There are two phone numbers in one cell.

Step 2: Convert to atomic values

Create separate rows:

SID	SName	PhoneNumber
S1	Ravi	9876
S1	Ravi	8765
S2	Priya	7654

Step 3: Identify the key

The combination:

(SID, PhoneNumber)

can uniquely identify each row.

Result

The relation is now in 1NF because every cell contains exactly one value.

1NF condition: No repeating groups or multi-valued attributes.

QUESTION : 3. Decompose R(A,B,C,D,E) into 2NF

Given:

R(A,B,C,D,E)

FDs:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

Step 1: Find candidate key

From:

$A \rightarrow B,C$

and

$BC \rightarrow D$

and

$D \rightarrow E$

we get:

A+ = {A,B,C,D,E}

Therefore:

A is a candidate key.

Step 2: Check 2NF

- 2NF eliminates partial dependencies of non-prime attributes on part of a composite candidate key.

- Here the candidate key is A, which contains only one attribute.

Therefore, a partial dependency is not possible.

So:

R is already in 2NF.

But there is a transitive dependency

We have:

$A \rightarrow BC$

$BC \rightarrow D$

$D \rightarrow E$

Therefore:

- $A \rightarrow D \rightarrow E$
- This is a transitive dependency and is handled when moving toward 3NF.

- 2NF result
- Since A is a single-attribute key:
- R(A,B,C,D,E) is already in 2NF.

If decomposition toward 3NF is required:

- R1(A,B,C)
- R2(B,C,D)
- R3(D,E)

QUESTION : 4. 3NF Decomposition of EMPLOYEE

Relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Assume:

- EmpID → EmpName, DeptID

- DeptID → DeptName, ManagerID

Functional Dependencies

EmpID → EmpName, DeptID

DeptID → DeptName, ManagerID

Therefore:

EmpID → DeptID → DeptName, ManagerID

This is a transitive dependency.

Step 1: Employee relation

Create:

EMPLOYEE(EmpID, EmpName, DeptID)

Primary key:

EmpID

Step 2: Department relation

Create:

DEPARTMENT(DeptID, DeptName, ManagerID)

Primary key:

DeptID

Final 3NF decomposition

EMPLOYEE(EmpID, EmpName, DeptID)

DEPARTMENT(DeptID, DeptName, ManagerID)

The transitive dependency has been removed.

QUESTION : 5 BCNF Verification and Decomposition

Relation:

R(A,B,C,D)

FDs:

- $AB \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

Step 1: Find candidate keys

AB+

$AB \rightarrow C$

$C \rightarrow D$

Therefore:

AB+ = {A,B,C,D}

So AB is a candidate key.

BC+

$C \rightarrow D$

$D \rightarrow A$

Therefore:

BC+ = {A,B,C,D}

So BC is a candidate key.

BD+

$D \rightarrow A$

But we cannot obtain C.

So BD is not a key.

Step 2: Check BCNF

For every non-trivial FD X → Y, X must be a superkey.

Consider:

C → D

C is not a superkey.

Therefore, the relation is not in BCNF.

Also:

D → A

D is not a superkey.

Therefore BCNF is violated.

Step 3: Decompose using C → D

Create:

R1(C,D)

and

R2(A,B,C)

Check R1

In R1:

C → D

C is a key of R1.

Therefore R1 is in BCNF.

Check R2

R2(A,B,C) has:

AB → C

AB is a key.

No BCNF violation.

Final BCNF decomposition

R1(C,D)

R2(A,B,C)

QUESTION : 6. Lossless-Join Decomposition Using Chase Algorithm

Consider:

R(A,B,C)

FD:

A → B

Decompose into:

- R1(A,B)
- R2(A,C)

We need to verify whether the decomposition is lossless.

Step 1: Create the tableau

Row	A	B	C
R1	a	b	c_1
R2	a	b_2	c

Attributes belonging to the relation are represented using common symbols.

Step 2: Apply FD

FD:

$A \rightarrow B$

Both rows have the same value of A.

Therefore, B values must become the same.

So:

Row	A	B	C
R1	a	b	c_1
R2	a	b	c

Step 3: Check result

A row contains all common symbols:

a, b, c

Therefore, the decomposition is lossless.

Conclusion

R(A,B,C) → R1(A,B), R2(A,C) is a lossless-join decomposition.

QUESTION : 7. Multivalued Dependencies and 4NF

Relation:

TEACHER(TID, Subject, Hobby)

Suppose a teacher can teach multiple subjects and has multiple hobbies, and these two sets are independent.

For example:

TID	Subject	Hobby
T1	DBMS	Music
T1	DBMS	Cricket

TID	Subject	Hobby
T1	OS	Music
T1	OS	Cricket

Multivalued Dependencies

We have:

TID →→ Subject

and

TID →→ Hobby

These are multivalued dependencies.

Problem

- Subjects and hobbies are independent.
- This creates unnecessary combinations and redundancy.

Decomposition

Create:

TEACHER_SUBJECT(TID, Subject)

TID	Subject
T1	DBMS
T1	OS

And:

TEACHER_HOBBY(TID, Hobby)

TID	Hobby
T1	Music
T1	Cricket

Final 4NF decomposition

- TEACHER_SUBJECT(TID, Subject)
- TEACHER_HOBBY(TID, Hobby)
- This removes the multivalued dependency from the original relation.

QUESTION : 8. Join Dependency and 5NF

Suppose we have:

SUPPLIER_PART_PROJECT(Supplier, Part, Project)

A supplier may supply different parts for different projects.

Example:

Supplier	Part	Project
S1	P1	J1
S1	P2	J1
S1	P1	J2
S1	P2	J2

The relation can contain redundant combinations.

Join Dependency

The relation may be represented by three relations:

SP(Supplier, Part)

SJ(Supplier, Project)

PJ(Part, Project)

Decomposition

Supplier-Part

SP(Supplier, Part)

Supplier-Project

SJ(Supplier, Project)

Part-Project

PJ(Part, Project)

The original relation can be reconstructed by joining these relations.

5NF

A relation is in 5NF (PJ/NF) when every non-trivial join dependency is implied by candidate keys.

Therefore, decomposing:

SUPPLIER_PART_PROJECT

into:

- SP(Supplier, Part)
- SJ(Supplier, Project)
- PJ(Part, Project)

helps eliminate join-based redundancy.

QUESTION : 9. Database Anomalies and Normalization to 3NF

Consider a poorly designed college relation:

COLLEGE(StudentID, StudentName, CourseID, CourseName, InstructorID, InstructorName)

Example:

StudentID	StudentName	CourseID	CourseName	InstructorID	InstructorName
S1	Ravi	C1	DBMS	I1	Kumar
S2	Priya	C1	DBMS	I1	Kumar

Functional Dependencies

StudentID → StudentName

CourseID → CourseName, InstructorID

InstructorID → InstructorName

Anomalies

1. Update anomaly

If instructor Kumar's name changes, it must be updated in multiple rows.

If one row is not updated, inconsistent data occurs.

2. Insertion anomaly

We cannot easily insert a new course if no student has enrolled in that course.

3. Deletion anomaly

If the last student enrolled in a course is deleted, information about that course and instructor may also be lost.

3NF Decomposition

STUDENT

STUDENT(StudentID, StudentName)

Primary key:

StudentID

COURSE

COURSE(CourseID, CourseName, InstructorID)

Primary key:

CourseID

INSTRUCTOR

INSTRUCTOR(InstructorID, InstructorName)

Primary key:

- InstructorID

- Final schema
- STUDENT(StudentID, StudentName)

- COURSE(CourseID, CourseName, InstructorID)

- INSTRUCTOR(InstructorID, InstructorName)
- This removes the major insertion, deletion and update anomalies.

QUESTION : 10. Attribute Closure and Candidate Keys

Attribute Closure

The closure of an attribute set X, written as:

X^+

is the set of all attributes that can be functionally determined by X using the given FDs.

Example

Consider:

R(A,B,C,D,E)

FDs:

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$

- D → E

Find A+.

Step 1

Start:

A+ = {A}

Using A → B:

A+ = {A,B}

Using B → C:

A+ = {A,B,C}

Using C → D:

A+ = {A,B,C,D}

Using D → E:

A+ = {A,B,C,D,E}

Therefore:

A+ = {A,B,C,D,E}

Since A determines all attributes:

A is a candidate key.

General method for finding candidate keys

1. Start with an attribute set X.
2. Calculate X+.
3. Apply all applicable FDs repeatedly.
4. If X+ contains all attributes, X is a superkey.
5. Check whether any attribute can be removed from X while still determining all attributes.
6. If no attribute can be removed, X is a candidate key.

Important exam point

Candidate key = minimal superkey.

For example, if:

$AB+ = \{A,B,C,D,E\}$

but:

$A+ \neq$ all attributes

and

$B+ \neq$ all attributes

then AB is a candidate key.